

Electrical energy and power

Whenever there is a current in a component, energy is transferred from the power source to that component.

Electrical power

The rate of energy transfer by each electrical component is called the electrical power. The equation for electrical power P in watts W is $P = VI$.

Using the equation $V = IR$, we get two additional equations for power:

$$P = I^2 R$$

$$P = \frac{V^2}{R}$$

Also,

$$P = \frac{E}{t}$$

$$V = \frac{E}{Q}$$

$$W = VQ$$

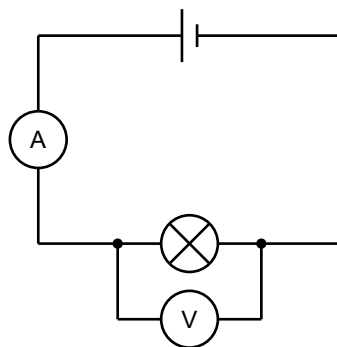
$$P = \frac{VQ}{t}$$

And

$$E = VIt$$

Summary Questions

1.



Using this circuit, measure the current of the circuit and the voltage across the component. Then, multiply these two values together to find the power of the filament lamp using the equation $P = VI$.

2.

$$P = VI$$

$$P = 8.0\text{V} \times 5.0\text{A}$$

$$= 40\text{W}$$

3.

$$\begin{aligned}\text{a.} \quad P &= VI \\ I &= \frac{P}{V} \\ &= \frac{1200\text{W}}{20\text{V}} \\ &= 60\text{A}\end{aligned}$$

$$\begin{aligned}\text{b. } W &= Pt \\ &= 1200 \times (1 \times 60 \times 60) \\ &= 4.3 \text{ MJ}\end{aligned}$$

4.

a. When R doubles, and I remains constant, V must halve because $R = \frac{V}{I}$.

Therefore the rate of energy transfer must halve, since $P = VI$.

b. When I doubles, but R remains unchanged, V must also double because $R = \frac{V}{I}$.

So the rate of energy transfer must quadruple because $P = VI$.

If $P_2 = (2V_1)(2I_1)$, $P_2 = 4V_1I_1$.

5.

$$\begin{aligned}[\text{W}] &= \left[\frac{\text{J}}{\text{s}} \right] \text{ because } E = VIt \text{ and } P = IV \\ &= \left[\frac{\text{kg m}^2\text{s}^{-2}}{\text{s}} \right] \\ &= [\text{kg m}^2\text{s}^{-3}]\end{aligned}$$

$$\begin{aligned}[\text{VA}] &= \frac{[\text{P}]}{[\text{I}]} \times [\text{A}] \text{ because } P = IV \\ &= \frac{[(\text{kg m}^2\text{s}^{-3})]}{[\text{A}]} [\text{A}] \text{ from our earlier calculation} \\ &= [\text{kg m}^2\text{s}^{-3}]\end{aligned}$$

So $1 \text{ W} = 1 \text{ VA}$.